Foundation for Advancement of Science & Technology

HW # 28

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(a)

i) input is 2D
output is 1D $\rightarrow$ $T : \mathbb{R}^2 \rightarrow \mathbb{R}^1$

ii) $\underset{\text{T.Matrix}}{m\times n}\ \underset{\text{input}}{2\times 1} = \underset{\text{output}}{1\times 1}$

n must be 2
m must be 1

1 row 2 columns

iii) $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\Rightarrow e_2 + 2e_1 \Rightarrow \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Horizontal Shear $= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Reflection across $y=x$ $= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ by $F_v = 2uu^T - I$

Rotation $= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

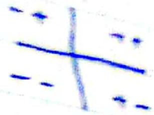
Transformation $= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$

$= \begin{bmatrix} -1 & -2 \\ 0 & 1 \end{bmatrix}$

CamScanner

Reflection x_1-axis $= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$



Reflection x_2-axis $= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Transformation $= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

rotation about origin is $\theta = 180°$

$\cos\theta = -1$
$\sin\theta = 0$ $\Rightarrow \theta = 180°$

(b)

i) $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix} x_1 + \begin{bmatrix} 9 \\ -7 \\ 3 \end{bmatrix} x_2$

~~T(A)~~ $A = \begin{bmatrix} -5 & 9 \\ 4 & -7 \\ 1 & 3 \end{bmatrix}$

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ii) $\left[\begin{array}{cc|c} -5 & 9 & 0 \\ 4 & -7 & 0 \\ 1 & 3 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & 0 \\ 4 & -7 & 0 \\ -5 & 9 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & -19 & 0 \\ 0 & 24 & 0 \end{array}\right] \sim$

$\left[\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$

$x_1 = x_2 = 0$

Ker(T) = 0 vector

T is one-to-one

iii) Span of T $= \left\{ \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 9 \\ -7 \\ 3 \end{bmatrix} \right\}$

It spans 2D subspace not entire $\mathbb{R}^3$ this is because there are only 2 independent vectors.

Range(T) = 2 therefore it does not occupy 3 that's why it is not onto. It is not invertible.